PRINCIPAL ENGINEER INTERVIEW PREP

3-Month Master Plan

Kashyap Pandey | 11.5 Years Java/Spring Boot | April 1 – June 30, 2026

Target: Microsoft • Google • Meta • Amazon

DSA (Trees→Graphs→DP→Patterns) + HLD + LLD + Mock Interviews

Prep Philosophy — Read This First

This is a complete fresh start. Every topic is taught from scratch using real-world examples before any code. No assumptions about prior knowledge. Depth over breadth — understanding WHY beats memorising WHAT every time.

Teaching Approach: Real example → Intuition → Step-by-step → Dry run → Code → Edge cases. Never jump to code without understanding first.

Principal Engineer Standard: For every problem: Correctness + Time/Space complexity + Trade-offs + Edge cases + Follow-up optimisation. Always explain WHY, not just WHAT.

- 240 total hours over 3 months — every hour planned
- Weekdays: 2 hrs/day (Mon–Fri) = 10 hrs/week
- Weekends: 5 hrs/day (Sat + Sun) = 10 hrs/week
- Total: ~20 hrs/week × 13 weeks = 240 hours
- Spaced repetition built into every week — no cramming
- Mock interviews every Sunday from Week 3 onwards

Weekly Schedule — At A Glance

Week	Dates	Topic	Hours	Start Command
MONTH 1 (Apr 1–Apr 30) — DSA FOUNDATIONS: Trees + Graphs + Backtracking				

W1	Apr 1–6	**Binary Trees — Structure, Traversals (Inorder/Preorder/Postorder), Height, Diameter**	14h	Mar 1 — Trees, let's begin
W2	Apr 7–13	**Binary Trees Advanced — LCA, Path Sum, Views, Symmetric, Zigzag BFS**	14h	Apr 7 — Trees Advanced, let's begin
W3	Apr 14–20	**BST — Insert/Delete/Search, Validate BST, Kth Smallest, Convert to sorted array + Mock #1**	14h	Apr 14 — BST, let's begin
W4	Apr 21–27	**Graphs — BFS, DFS, Cycle Detection, Topological Sort, Connected Components**	14h	Apr 21 — Graphs, let's begin
W4+	Apr 28–30	**Graphs Advanced — Dijkstra, Bellman-Ford, Union Find + Mock #2**	6h	Apr 28 — Graphs Advanced, let's begin
MONTH 2 (May 1–May 31) — DP + ADVANCED PATTERNS + LLD				
W5	May 1–4	**Backtracking — Permutations, Combinations, Subsets, N-Queens, Sudoku**	8h	May 1 — Backtracking, let's begin
W6	May 5–11	**Dynamic Programming 1D — Fibonacci, Climbing Stairs, House Robber, Coin Change**	14h	May 5 — DP 1D, let's begin
W7	May 12–18	**Dynamic Programming 2D — Grid paths, LCS, Edit Distance, 0/1 Knapsack + Mock #3**	14h	May 12 — DP 2D, let's begin
W8	May 19–25	**Heaps + Priority Queue — Top K, Merge K Lists, Median Stream, Task Scheduler**	14h	May 19 — Heaps, let's begin
W9	May 26–31	**LLD Week 1 — OOP Refresh, SOLID, Design Patterns + LLD: Parking Lot, Elevator + Mock #4**	12h	May 26 — LLD, let's begin
MONTH 3 (Jun 1–Jun 30) — PATTERNS + SYSTEM DESIGN + MOCK LOOPS				
W10	Jun 1–7	**Core Patterns — Sliding Window, Two Pointers, Binary Search, Prefix Sum, Stack**	14h	Jun 1 — Core Patterns, let's begin
W11	Jun 8–14	**Advanced Patterns — Tries, Intervals, Monotonic Stack, Bit Manipulation + Mock #5**	14h	Jun 8 — Advanced Patterns, let's begin
W12	Jun 15–21	**LLD Week 2 — Machine Coding: Splitwise, BookMyShow, Snake Game + Mock #6**	14h	Jun 15 — LLD Machine Coding, let's begin
W13	Jun 22–28	**Full Mock Loop Week — 2 mocks/day, all patterns, weak area drill**	14h	Jun 22 — Mock Loop, let's begin

W14	Jun 29–30	**Final Polish — Company specific, behavioral stories, last review**	4h	Jun 29 — Final Polish, let's begin

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System Design Saturdays — Full Schedule

Approach: Every Saturday = 5hrs. HLD first (3hrs) then LLD (2hrs). Same teaching pattern: real example → components → deep dive → trade-offs → 30-second answer.

Sat	Design	HLD + LLD Focus
Apr 5	**HLD Foundations**	What is distributed system, CAP theorem, consistency models, 8 building blocks, HLD template (7 steps)
Apr 12	**Rate Limiter**	HLD: algorithms (Fixed/Sliding/Token/Leaky), Redis, distributed counting. LLD: RateLimiter class design
Apr 19	**URL Shortener**	HLD: hashing, DB design, redirect flow. LLD: ShortenerService, URLMapper class design
Apr 26	**Notification System**	HLD: push/email/SMS, fan-out, queues. LLD: NotificationService, Template pattern
May 3	**Key-Value Store**	HLD: consistent hashing, replication, CAP. LLD: Storage engine interface, LSM tree concept
May 10	**Design Twitter Feed**	HLD: fan-out on write vs read, timeline service. LLD: Tweet, User, Feed classes
May 17	**Design Uber**	HLD: geo-indexing, ride matching, surge pricing. LLD: Trip, Driver, Rider, MatchingEngine
May 24	**Design Kafka**	HLD: topics, partitions, consumers, offsets. LLD: Producer/Consumer interface, partition strategy
May 31	**Design Payment System**	HLD: idempotency, retries, ledger, fraud hooks. LLD: Transaction, PaymentProcessor, Retry pattern
Jun 7	**Design Google Search**	HLD: crawling, indexing, ranking, caching. LLD: Crawler, Indexer, RankEngine interfaces
Jun 14	**Design WhatsApp**	HLD: messaging, presence, media, E2E encrypt. LLD: Message, Chat, DeliveryReceipt classes
Jun 21	**Design Amazon**	HLD: product catalog, cart, order, inventory. LLD: Order, Inventory, Cart, Checkout classes
Jun 28	**Full Mock System Design**	Full 45-min HLD+LLD mock — interviewer style, no hints, full debrief

💻 DSA Detailed Plan — All Patterns + Problems

Problem Selection: Based on NeetCode 150 + Blind 75 + MAANG interview reports 2025-2026. Every problem selected for frequency at Microsoft, Google, Meta, Amazon.

MONTH 1 — TREES

Why Trees First?

Trees build recursion intuition which is the foundation of Graphs AND Dynamic Programming. Master trees first = everything else gets easier. Every MAANG company asks 2-3 tree questions in every interview loop.

Wk	Pattern / Topic	Key Problems (LeetCode)	MAANG Frequency
Week 1 (Apr 1–6) — Binary Tree Basics + Traversals			
Mon	**Tree Structure & Node**	Understand Node class, left/right, null. Build tree manually	Foundation
Mon	**Inorder Traversal**	94. Binary Tree Inorder Traversal (iterative + recursive)	★★★★★
Tue	**Preorder + Postorder**	144. Preorder, 145. Postorder (both iterative + recursive)	★★★★★
Wed	**Level Order BFS**	102. Binary Tree Level Order Traversal (queue based)	★★★★★
Thu	**Tree Height + Depth**	104. Maximum Depth, 111. Minimum Depth	★★★★
Fri	**Tree Diameter**	543. Diameter of Binary Tree	★★★★★
Sat	**System Design**	HLD Foundations — CAP, 8 blocks, HLD template (5 hrs)	—
Sun	**Mock #0 (Warm-up)**	Inorder + Level Order — timed 30 min each	Confidence
Week 2 (Apr 7–13) — Binary Tree Advanced			
Mon	**Invert Tree**	226. Invert Binary Tree	★★★★★
Mon	**Symmetric Tree**	101. Symmetric Tree	★★★★
Tue	**Path Sum**	112. Path Sum, 113. Path Sum II (all paths)	★★★★★
Wed	**LCA**	236. Lowest Common Ancestor of Binary Tree	★★★★★
Thu	**Right/Left View**	199. Binary Tree Right Side View	★★★★★
Thu	**Zigzag Traversal**	103. Binary Tree Zigzag Level Order	★★★★
Fri	**Max Path Sum**	124. Binary Tree Maximum Path Sum (Hard)	★★★★★
Fri	**Flatten to Linked List**	114. Flatten Binary Tree to Linked List	★★★★

Sat	System Design	Rate Limiter — HLD + LLD (5 hrs)	—
Sun	Mock #1	2 tree problems timed (35 min each) + full debrief	Assessment
Week 3 (Apr 14–20) — BST + Trie Intro			
Mon	BST Insert/Search/Delete	700. Search BST, 701. Insert, 450. Delete Node	★★★★
Tue	Validate BST	98. Validate Binary Search Tree	★★★★★
Tue	Kth Smallest	230. Kth Smallest Element in BST	★★★★★
Wed	BST to Sorted Array	538. Convert BST to Greater Tree, 108. Sorted Array to BST	★★★★
Wed	LCA in BST	235. LCA of BST (simpler than binary tree)	★★★★
Thu	Trie — Intro	208. Implement Trie (Insert, Search, StartsWith)	★★★★★
Fri	Word Search II	212. Word Search II (Trie + Backtracking)	★★★★
Sat	System Design	URL Shortener — HLD + LLD (5 hrs)	—
Sun	Mock #2	BST problem + Tree Hard — timed + debrief	Assessment

MONTH 1 (cont) — GRAPHS

Why Graphs After Trees?

Trees are graphs with no cycles. Moving from trees to graphs = extending your existing mental model. BFS on trees = BFS on graphs. DFS on trees = DFS on graphs. The transition is natural.

Wk	Pattern / Topic	Key Problems (LeetCode)	MAANG Frequency
Week 4 (Apr 21–27) — Graphs BFS + DFS + Cycle Detection			
Mon	Graph Representation	Adjacency list vs matrix, build graph from edge list	Foundation
Mon	BFS on Graph	Number of Islands (200) — BFS approach	★★★★★
Tue	DFS on Graph	Number of Islands (200) — DFS approach, Clone Graph (133)	★★★★★
Wed	Connected Components	Number of Connected Components (323), Max Area Island (695)	★★★★
Wed	Cycle Detection	207. Course Schedule (detect cycle in directed graph)	★★★★★
Thu	Topological Sort	210. Course Schedule II (topo sort BFS — Kahn's algorithm)	★★★★★
Thu	Topological Sort DFS	Topo sort using DFS + visited/in-stack approach	★★★★

Fri	Flood Fill + Matrix BFS	733. Flood Fill, 994. Rotting Oranges (multi-source BFS)	★★★★★
Sat	System Design	Notification System — HLD + LLD (5 hrs)	—
Sun	Spaced Rep + Practice	BFS/DFS problems mixed — identify pattern, time yourself	Drill
Week 4+ (Apr 28–30) — Advanced Graph Algorithms			
Mon	Union Find	547. Number of Provinces, 684. Redundant Connection	★★★★★
Tue	Dijkstra's Algorithm	743. Network Delay Time, 1631. Min Effort Path	★★★★★
Wed	Bellman-Ford + A*	787. Cheapest Flights K Stops, concept of negative weights	★★★
Wed+	Mock #3	Graph problems — 2 timed + full debrief	Assessment

MONTH 2 — BACKTRACKING + DP + HEAPS

Wk	Pattern / Topic	Key Problems (LeetCode)	MAANG Frequency
Week 5 (May 1–4) — Backtracking			
Thu	Backtracking Template	Decision tree mental model, base case, explore, undo	Foundation
Thu	Subsets	78. Subsets, 90. Subsets II (with duplicates)	★★★★
Fri	Permutations	46. Permutations, 47. Permutations II	★★★★★
Fri	Combinations	77. Combinations, 39. Combination Sum, 40. Combination Sum II	★★★★★
Sat	System Design	Key-Value Store — HLD + LLD (5 hrs)	—
Sun	Backtracking Hard	51. N-Queens, 37. Sudoku Solver	★★★★
Week 6 (May 5–11) — Dynamic Programming 1D			
Mon	DP Introduction	What is DP, memoization vs tabulation, overlapping subproblems	Foundation
Mon	Fibonacci / Climbing	70. Climbing Stairs, 509. Fibonacci Number	★★★★
Tue	House Robber	198. House Robber, 213. House Robber II (circular)	★★★★★
Wed	Coin Change	322. Coin Change, 518. Coin Change II	★★★★★
Wed	Min Cost Climbing	746. Min Cost Climbing Stairs	★★★★
Thu	Jump Game	55. Jump Game, 45. Jump Game II	★★★★★
Thu	Decode Ways	91. Decode Ways	★★★★

Fri	**Word Break**	139. Word Break	★★★★★
Fri	**Longest Increasing Subseq**	300. LIS (DP + binary search approach)	★★★★★
Sat	**System Design**	Twitter Feed — HLD + LLD (5 hrs)	—
Sun	**Mock #4**	DP 1D problem + Backtracking problem — timed + debrief	Assessment
Week 7 (May 12–18) — Dynamic Programming 2D			
Mon	**Grid DP**	62. Unique Paths, 63. Unique Paths II, 64. Min Path Sum	★★★★★
Tue	**0/1 Knapsack**	Classic knapsack — taught from scratch with real example	★★★★★
Tue	**Target Sum**	416. Partition Equal Subset Sum, 494. Target Sum	★★★★★
Wed	**LCS**	1143. Longest Common Subsequence	★★★★★
Wed	**Edit Distance**	72. Edit Distance (Hard — classic 2D DP)	★★★★★
Thu	**Longest Common Substr**	718. Maximum Length of Repeated Subarray	★★★★
Thu	**Distinct Subsequences**	115. Distinct Subsequences (Hard)	★★★
Fri	**Interleaving String**	97. Interleaving String	★★★★
Sat	**System Design**	Uber — HLD + LLD (5 hrs)	—
Sun	**Mock #5**	DP 2D problem timed + system design verbal — debrief	Assessment
Week 8 (May 19–25) — Heaps + Priority Queue			
Mon	**Heap Introduction**	Min heap vs max heap, PriorityQueue in Java, heapify	Foundation
Mon	**Kth Largest**	215. Kth Largest Element in Array	★★★★★
Tue	**Top K Frequent**	347. Top K Frequent Elements	★★★★★
Tue	**K Closest Points**	973. K Closest Points to Origin	★★★★
Wed	**Merge K Sorted Lists**	23. Merge K Sorted Lists (Hard)	★★★★★
Wed	**Task Scheduler**	621. Task Scheduler	★★★★
Thu	**Find Median Stream**	295. Find Median from Data Stream (Hard — 2 heaps)	★★★★★
Thu	**Sliding Window Maximum**	239. Sliding Window Maximum (Monotonic Deque)	★★★★★
Fri	**Meeting Rooms**	253. Meeting Rooms II (intervals + heap)	★★★★★
Sat	**System Design**	Kafka — HLD + LLD (5 hrs)	—
Sun	**Spaced Rep + Drill**	Heap problems mixed — identify pattern, time yourself	Drill

Week 9 (May 26–31) — LLD Week 1			
Mon	**OOP Refresh**	Encapsulation, Inheritance, Polymorphism, Abstraction — real examples	Foundation
Mon	**SOLID Principles**	Each principle with real Java examples and violations	Foundation
Tue	**Design Patterns**	Factory, Singleton, Observer, Strategy, Decorator — when to use	★★★★★
Wed	**LLD: Parking Lot**	Full class design — ParkingLot, Spot, Vehicle, Ticket, Payment	★★★★★
Thu	**LLD: Elevator System**	ElevatorController, Elevator, Request, Scheduling algorithm	★★★★
Fri	**LLD: Library Management**	Book, Member, Loan, Reservation, Search — full design	★★★★
Sat	**System Design**	Payment System — HLD + LLD (5 hrs)	—
Sun	**Mock #6**	LLD problem timed 45 min (Parking Lot or Elevator)	Assessment

MONTH 3 — CORE PATTERNS + ADVANCED + MACHINE CODING

Wk	Pattern / Topic	Key Problems (LeetCode)	MAANG Frequency
Week 10 (Jun 1–7) — Core Patterns (Sliding Window + Two Pointers + Binary Search + Stack)			
Mon	**Sliding Window**	3. Longest Substring No Repeat, 76. Min Window Substring, 239. Sliding Max	★★★★★
Tue	**Two Pointers**	15. 3Sum, 11. Container Water, 42. Trapping Rain Water (Hard)	★★★★★
Wed	**Binary Search**	33. Rotated Array, 153. Find Min Rotated, 4. Median of Two Sorted Arrays (Hard)	★★★★★
Thu	**Stack — Monotonic**	84. Largest Rectangle Histogram, 85. Maximal Rectangle (Hard)	★★★★★
Thu	**Stack Problems**	20. Valid Parentheses, 155. Min Stack, 739. Daily Temperatures	★★★★★
Fri	**Prefix Sum**	560. Subarray Sum=K, 238. Product Except Self, 304. Range Sum Query 2D	★★★★★
Sat	**System Design**	Google Search — HLD + LLD (5 hrs)	—
Sun	**Mock #7**	Mixed patterns — 2 problems timed + debrief	Assessment
Week 11 (Jun 8–14) — Advanced Patterns			
Mon	**Intervals**	56. Merge Intervals, 57. Insert Interval, 252. Meeting Rooms	★★★★★
Tue	**Greedy**	55. Jump Game, 135. Candy, 435. Non-overlapping Intervals	★★★★

Wed	**Bit Manipulation**	191. Number of 1 Bits, 338. Counting Bits, 268. Missing Number	★★★
Thu	**Linked List**	206. Reverse, 21. Merge Two Lists, 141. Cycle, 23. Merge K Lists	★★★★★
Thu	**Advanced LinkedList**	25. Reverse K Group (Hard), 138. Copy List Random Pointer	★★★★★
Fri	**String Problems**	5. Longest Palindromic Substring, 647. Palindromic Substrings	★★★★★
Sat	**System Design**	WhatsApp — HLD + LLD (5 hrs)	—
Sun	**Mock #8**	Advanced patterns — 2 problems timed + debrief	Assessment
Week 12 (Jun 15–21) — LLD Week 2 — Machine Coding			
Mon	**Machine Coding Intro**	What is machine coding round, how to approach 90 min problem	Foundation
Mon	**LLD: Splitwise**	User, Group, Expense, Balance, Settlement — full working code	★★★★★
Tue	**LLD: BookMyShow**	Show, Screen, Seat, Booking, Payment — full working code	★★★★★
Wed	**LLD: Snake and Ladder**	Board, Player, Snake, Ladder, Dice — full working code	★★★★
Thu	**LLD: Cab Booking**	Driver, Rider, Trip, Matching, Pricing — full working code	★★★★★
Fri	**LLD: Food Delivery**	Restaurant, Order, DeliveryAgent, Tracking — full working code	★★★★
Sat	**System Design**	Amazon — HLD + LLD (5 hrs)	—
Sun	**Mock #9**	Machine coding — 90 min timed (Splitwise or BookMyShow)	Assessment
Week 13 (Jun 22–28) — Full Mock Loop Week			
Mon	**Mock Loop 1**	DSA Hard (Trees/Graphs) — 2 problems 35 min each	Full Loop
Tue	**Mock Loop 2**	DSA Hard (DP) + System Design verbal (45 min)	Full Loop
Wed	**Mock Loop 3**	Machine Coding 90 min (unseen LLD problem)	Full Loop
Thu	**Mock Loop 4**	Full interview simulation — DSA + SD + Behavioral (3 hrs)	Full Loop
Fri	**Weak Area Drill**	Identify gaps from mock loops — targeted 2 hr drill	Repair
Sat	**System Design**	Full Mock System Design 45 min — no hints	Full Loop
Sun	**Mock Loop 5**	Final full loop — company specific (Google/Meta/Amazon/MS)	Final
Week 14 (Jun 29–30) — Final Polish			

Mon	**Company Specific**	Microsoft: OOP + DP. Google: Hard DP + Graphs. Meta: Graphs + Trees. Amazon: DP + System Design	Final
Tue	**Behavioral Stories**	STAR format — 5 stories ready: leadership, conflict, failure, scale, influence	Final

🎯 Monthly Milestones — How You Know You're On Track

Month	You Can DO This	You Can SAY This
April End	• Solve any tree/graph medium in <25 min • Write BFS/DFS without template • Design Rate Limiter + URL Shortener end-to-end	• Explain why DFS vs BFS for any problem • State 3 trade-offs for any system design
May End	• Solve DP medium with memoization + tabulation • Complete LLD problem (Parking Lot) in 45 min • Design Uber + Kafka + Payment end-to-end	• Explain DP recurrence relation clearly • Walk through SOLID violations in real code
June End	• Pass full mock loop (DSA + LLD + HLD) in 3 hrs • Machine code Splitwise/BookMyShow in 90 min • Solve Hard DP / Graph in 40 min	• Behavioral: 5 STAR stories fluently • Company-specific: tailor answers per company

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▶ Session Start Commands

Say exactly these commands to start each session. This keeps context tight and focused.

DSA Sessions	System Design Sessions
`Today is Apr 1 — Trees Basics, let's begin`	`Today is Apr 5 — HLD Foundations, let's begin`
`Today is Apr 7 — Trees Advanced, let's begin`	`Today is Apr 12 — Rate Limiter, let's begin`
`Today is Apr 14 — BST, let's begin`	`Today is Apr 19 — URL Shortener, let's begin`
`Today is Apr 21 — Graphs BFS/DFS, let's begin`	`Today is Apr 26 — Notification System, let's begin`
`Today is Apr 28 — Graphs Advanced, let's begin`	`Today is May 3 — Key-Value Store, let's begin`
`Today is May 1 — Backtracking, let's begin`	`Today is May 10 — Twitter Feed, let's begin`

Today is May 5 — DP 1D, let's begin	Today is May 17 — Uber, let's begin
Today is May 12 — DP 2D, let's begin	Today is May 24 — Kafka, let's begin
Today is May 19 — Heaps, let's begin	Today is May 31 — Payment System, let's begin
Today is May 26 — LLD Week 1, let's begin	Today is Jun 7 — Google Search, let's begin
Today is Jun 1 — Core Patterns, let's begin	Today is Jun 14 — WhatsApp, let's begin
Today is Jun 8 — Advanced Patterns, let's begin	Today is Jun 21 — Amazon, let's begin
Today is Jun 15 — LLD Machine Coding, let's begin	Today is Jun 28 — Final Mock SD, let's begin
Today is Jun 22 — Mock Loop Week, let's begin	
Today is Jun 29 — Final Polish, let's begin	

📋 Ground Rules — Non-Negotiable

- **Always say start command exactly — keeps context focused and on-plan**
- Spaced repetition every session — 10 min at start, no exceptions
- Pattern identification BEFORE writing any code — always
- Dry run BEFORE submitting code — always
- After every session: document updated, problems committed to GitHub
- If stuck >10 min: say 'hint please' — not staring at blank screen
- Mock interview rules: think out loud always, no silent coding
- If falling behind plan: discuss first, then adjust — never skip silently

"The goal is not to prepare for interviews. The goal is to become the engineer who deserves the role."